

Mohsen Dirbaz

Address: Chicago, US

PROFILE

A well-rounded Chemical Engineer with 5+ years in-depth knowledge of process design, modeling, and simulation in oil & gas industries, ASPEN Plus, HYSYS, PFD's, P&ID's, and heat and material balances. Strong foundation in process engineering with hands-on experience in core unit operations (e.g., distillation, heat exchange, reactors) and supporting equipment such as pumps, compressors, and separation systems. A solid combination of academic excellence, technical proficiency, and practical project experience, complemented by a proactive, self-motivated, and detail-oriented approach to individual and collaborative endeavors. Served as a Process Engineer at L&T Technology Services, where responsibilities included review of PFDs and PIDs, fluid flow calculations, and coordination with equipment vendors and fabrication partners. Technical capabilities span advanced simulation and programming tools such as Aspen Plus, HYSYS, MATLAB, Mathematica, Python, JavaScript, NumPy, and Jupyter notebooks. Recognized with multiple honors and awards, including Teaching Assistant of the Year at Illinois Institute of Technology and competitive graduate assistantships. Active participation in professional organizations such as AIChE and SPE, as well as leadership roles in academic associations.

EDUCATION

Chemical Engineering | Ph.D. Illinois Institute of Technology (IIT)

2012 - 2020 CHICAGO, US

GPA: 3.6 (2012-2020)

Dissertation: "A Neural Network Based Model for Biomass Gasification in Fluidized bed" Researched gasification and emission control technologies relevant to IGCC systems, including biomass and coal gasification in fixed, entrained, and fluidized bed reactors. Investigated intermediate purification and treatment steps—such as sulfur compound removal, trace metal abatement, COS hydrolysis, and carbon capture and other enabling technologies that enhance technical and economic viability of power generation and chemicals production. Conducted research on the environmental impacts of hydraulic fracturing and evaluated emerging frac water remediation strategies—including physico-chemical and biological treatment pathways, process models (e.g. water chemistry models for calculation of PH and precipitation reactions, mass transfer models for oxygen modeling and other gas/liquid interactions) with a focus on sustainable water management practices.

Natural Gas Engineering | M.S. Texas A&M University-Kingsville (TAMUK)

2010 - 2012 KINGSVILLE, US

GPA: 3.9 (2010-2012)

Dissertation: "Evaluation of Formation Damage in Horizontally Fractured Wells in Low Permeability Reservoirs" Performed advanced process simulation and optimization of industrial gas processing systems using HYSYS software, including acid gas sweetening, dehydration, and fractionation units. Designed and optimized compressor station configurations for a 1,800-mile natural gas transmission pipeline to ensure efficient flow and minimize pressure losses. Performed preliminary economic analysis and material selection to determine optimal pipe diameter, grade, and specification based on maximum pressure and performance requirements. Conducted specialized research in offshore petroleum engineering, focusing on subsea well completion technologies, cementitious material rheology, and enhanced oil recovery screening methodologies.

EDUCATION

Petroleum Engineering | B.Sc. Sharif University of Technology (SUT)

2005 - 2010 TEHRAN, IRAN
(2005-2010)

Gained hands-on experience in PVT lab, drilling lab, reservoir rock, and fluid mechanics lab. Completed a comparative analysis for constant composition expansion and constant volume depletion.

EXPERIENCE

Postdoctoral Fellow University of Missouri-Columbia

2023 - 2025 COLUMBIA, US

- Developed TEA Space, a platform that facilitates comprehensive techno-economic assessments by integrating dynamic modeling and simulation capabilities in a highly customizable interface.
- Conducted in-depth analysis on feasibility of manufacturing Nitrogen trifluoride, a critical material used by semiconductor foundries for wafer etching and chamber cleaning.
- Presented at AIChE: "Optimally Dispersed Network of Biomass Gasification to Produce Green H₂"
- Authored several technical reports:
 - NH₃ Report:
 - Gray Haber-Bosch Process. Advanced Blue Processes.
 - Carbon Capture Technologies. Hydrogen Transportation and Storage Options.
 - Green Ammonia via Electrochemical Synthesis. Lithium Mediated Ammonia Production.
- H₂ Report:
- Hydrogen Production Routes. Electrolysis. Fossil Resources. Biomass.
- Hydrogen Powered Fleet. Expansion Strategies and Market Penetration.
- Biomass Carbon Neutrality. Circular Economy. Life Cycle Assessment.
- Tinkered An Energy-Policy Roadmap, Introducing A Forward-Thinking Renewable Energy Blueprint Built on 6 Pillars of:
 - Climate liability (adaptation vs mitigation)
 - Electrifying rural economies (tribal, local, state and territorial models)
 - Sustainable forestry (management, certification, carbon accounting)
 - Biomass utilization (value chain pathway optimization)
 - Hydrogen demand (incentive mechanisms)

Energy Initiatives Consultancy Didas International Group/Iran

2020 - 2023 TEHRAN, IRAN

- Reviewed Regulatory Compliance Status of Flare Treatment Technologies.
- Mapped Silica Processing End Use Applications of Reflective Coatings for Solar Thermal Systems.
- Analyzed Impact of Euro Adoption on the GDP Dynamics of Middle Eastern Countries.
- Investigated Systemic Disparity of Training-Deployment in Engineering Education in US.

Process Engineer Larsen and Toubro Technology Services Limited

2019 - 2019 MUMBAI, INDIA

- Acted as key liaison between offshore engineers and clients, guaranteeing project alignment.
- Reviewed PFDs (Process Flow Diagrams) and P&IDs (Process and Instrumentation Diagrams).
- Delivered comprehensive progress reports at various milestones, fostering transparency and client trust.
- Collaborated with suppliers to finalize equipment designs, optimizing project outcomes.

EXPERIENCE

Teaching Assistant Illinois Institute of Technology

2015 - 2017 CHICAGO, US

- Served as Teaching Assistant in Fluid Mechanics, Thermodynamics I/II, and Process design I/II.

Intern Didas International Group/Iran

2009 - 2010 TEHRAN, IRAN

- Prepared technical reports and presented key insights to internal and external stakeholders.

HONORS AND AWARDS

Teaching Assistant of the Year Award at IIT

2016

Recipient of Teaching Assistant of the Year Award at IIT (2016 & 2017).

Graduate Assistantship, Illinois Institute of Technology

2012

Recipient of Graduate Assistantship, Illinois Institute of Technology, Fall 2012.

Graduate Assistantship, University of Texas A&M-Kingsville

2010

Recipient of Graduate Assistantship, University of Texas A&M-Kingsville, Fall 2010.

Full Scholarship, Sharif University of Technology

2005

Recipient of Full Scholarship, Sharif University of Technology, Fall 2005.

AFFILIATION

- Member of American Institute of Chemical Engineers (2014-present).
- Liaison of ChBE Ethics Committee (NSF-funded project, 2018)
- President of Graduate Student Association at ChBE department, IIT (2016-2017).
- Member of Society of Petroleum Engineers (2010-2012).
- Member of executive student committee of the 2nd and 3rd HSE conference at SUT (2009-2010).
- Representative of petroleum engineering students at Sharif University of Technology (2005-2010).